SAI NATH UNIVERSITY
JIRAWAR ORMANJHI, RANCHI

CLOUD BASED RESULT MANAGEMENT

A synopsis submitted in partial fulfillment of the requirement for the award of the degree

OF

Bachelor of Computer Application (BCA)

<u>Synopsis</u>

Submitted By:

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DECLARATION

We hereby declare that the project titled **“Cloud Based Result Management”** is our own work done as a part of our academic project. This project is completed by our group of three members under the guidance of our teachers.

The main aim of this project is to create a cloud based system that allows students to see their result easily and safely through the internet. This system helps to reduce physical inconvenience, saves time, and keeps proper records of result’s data. All the work related to deployment, architecture and preparing the project report has been done by us as a team.

We have used our own ideas and knowledge while developing this project. We take full responsibility for the originality of this project.

We understand that submitting false work is against academic rules. We confirm that the details given in this project are true to the best of our knowledge.

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Group Members:

1. **Name:** Adnan Amir **Roll No:** BCA23044
2. **Name:** Aatish Rana **Roll No:** BCA23110
3. **Name:** Nitin Kumar Mahli **Roll No:** BCA23071

CERTIFICATE

This is to certify that this is a bonafide record of the project work done satisfactory at **“BRIZTECH INFOSYSTEMS PVT. LTD”** by Adnan Amir, Aatish Rana, Nitin Kumar Mahli, in partial fulfillment of BCA Examination.

This report or similar report on the topic has not been submitted for any other examination and doesn’t form part of any other course undergone by the candidates.

External Signature **Internal Signature**

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1. Title of the Project

Cloud-Based Result Management

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2. Introduction

Educational institutions generate large amounts of student assessment data every semester. Traditional result processing systems are often manual or locally hosted, limiting accessibility, scalability, and security. To address these issues, a *Cloud-Based Result Management System* is proposed, enabling institutions to publish, manage, and store results on cloud platforms. Students and faculty can view results securely from anywhere, at any time, using internet connectivity. The system takes advantage of cloud capabilities such as scalability, storage reliability, high availability, and reduced infrastructure cost.

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3. Objectives

The objectives of the project include:

- To store student result data securely in the cloud.
- To provide 24×7 remote access for students and faculty.
- To eliminate manual result processing delays.
- To reduce infrastructure cost through cloud deployment.
- To ensure data backup and disaster recovery through cloud storage.
- To support scalable handling of large result datasets.

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4. Problem Statement

Manual and on-premise result management systems suffer from:

- Time-consuming result publication
- Limited accessibility (only within the institute premises)
- Higher chances of data loss or errors
- High infrastructure and maintenance cost

- Poor scalability for large institutions

Therefore, a cloud-based result management solution is required to ensure wider accessibility, improved performance, and enhanced data security.

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5. Scope of the Project

In-Scope:

- Cloud computing platform for hosting application and database
- Role-based login for students, faculty, and admin
- Online result retrieval by students
- Cloud storage for records, transcripts, and result archives
- Automatic data backup and recovery
- Real-time accessibility using internet

Out of Scope / Future Enhancements:

- Mobile native applications
- AI-based student performance analytics
- Integration with national academic platforms
- Proctoring or examination module

Target Users:

Universities, Colleges, Schools, Examination Boards

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6. System Requirements

Software Requirements

- Operating System: Windows / Linux
- Cloud Platform: (Any one)
  - AWS
  - Google Cloud Platform (GCP)
  - Microsoft Azure
  - IBM Cloud
- Cloud Services May Include:

  - **Compute:** EC2 / VM Instances / Serverless Functions
  - **Storage:** S3 / Blob Storage / Cloud Storage Buckets
  - **Database:** Cloud SQL / NoSQL DB / Managed DB Services
  - **Identity Access:** IAM / Cloud Authentication
  - **Hosting:** Cloud App Hosting Services

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: 4GB or higher
- Storage: 20GB minimum
- Network: Stable Internet Connectivity

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7. Methodology / Proposed System

The system is based on **Cloud Deployment Architecture**:

Client (Browser)
↓
Cloud Application Layer (SaaS)
↓
Cloud Database & Cloud Storage
↓
Cloud Infrastructure (IaaS / PaaS)

Cloud Computing Models Used

- **SaaS:** Students access result portal online.
- **PaaS:** Application runs on cloud-managed runtime environment.
- **IaaS:** Cloud resources for compute, storage & networking.

Cloud Benefits Applied

- Elastic Scalability
- Pay-as-you-go Billing
- High Availability
- Automatic Backup & Disaster Recovery
- Data Security & Access Control

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8. Modules of the System

1. **Authentication Module**
   - Cloud-based access control for users
2. **Admin Module**
   - Academic year, course & student management
3. **Faculty Module**
   - Marks entry & result uploading
4. **Result Processing Module**
   - Grade computation & cloud storage
5. **Student Module**
   - Online result viewing & downloading
6. **Cloud Storage Module**
   - Result archival & secure data retrieval
7. **Backup & Recovery Module**
   - Automatic cloud-based data backup

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9. Expected Outcomes

The project is expected to:

- Improve accessibility of student result records
- Reduce dependency on physical infrastructure
- Enhance data security & reliability
- Provide online result publishing without delays
- Decrease operational cost of IT resources
- Enable scalable academic data management

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10. Conclusion

The cloud-based result management system provides an effective digital solution for educational institutions by leveraging cloud capabilities. It improves accessibility, reduces cost, and ensures secure data handling while eliminating the drawbacks of traditional on-premise systems. The system supports future scalability and aligns with the digital transformation goals of modern educational environments.

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11. References

- IEEE Journals on Cloud Computing in Education
- Amazon Web Services (AWS) Documentation
- Google Cloud Platform (GCP) Documentation
- Microsoft Azure Documentation
- NIST Cloud Computing Reference Model
- MDN & Research Articles on Academic Data Management